

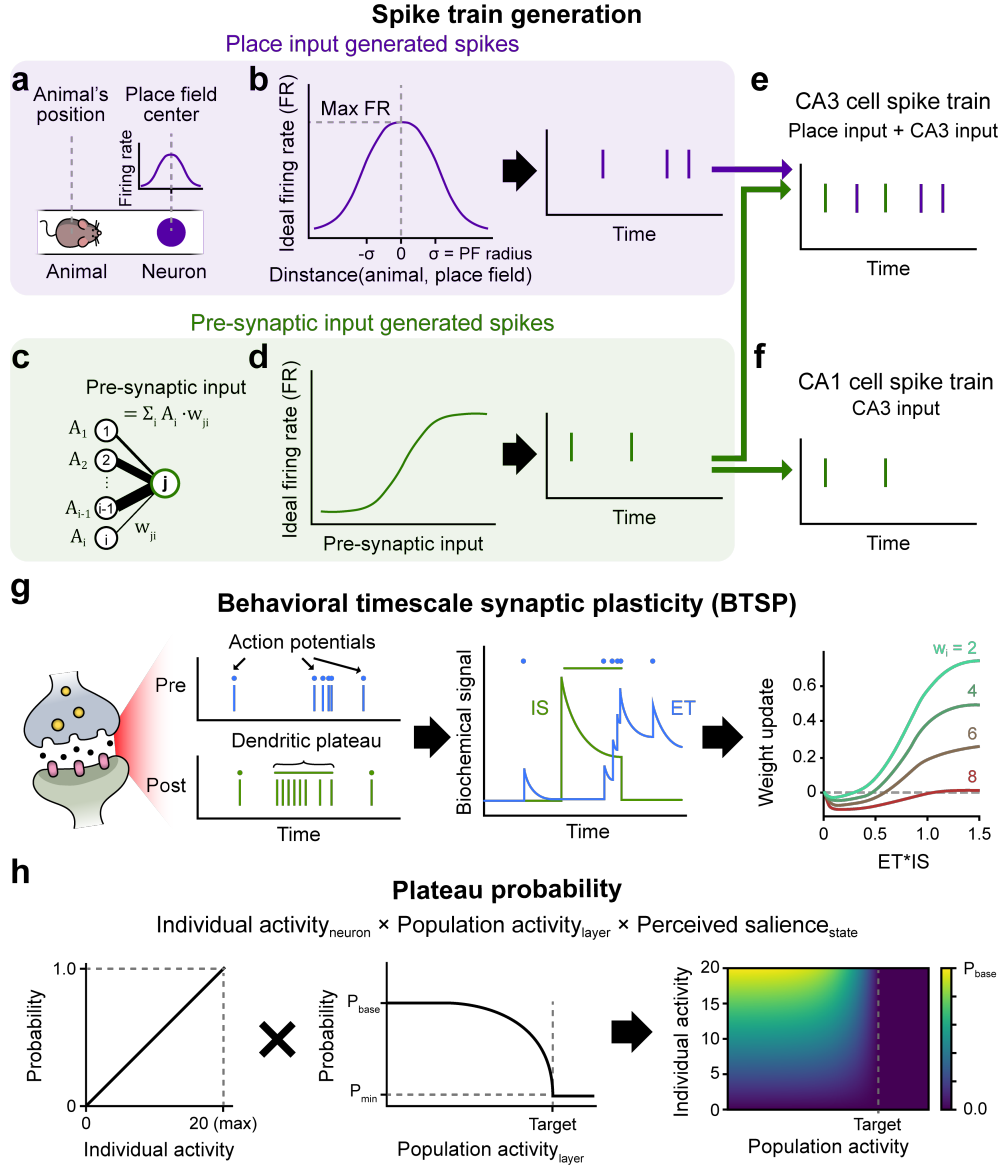
Supplementary materials

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1 Details of synaptic learning rule during the online phase



Supplementary Figure 1: Spiking neural network model activity. **a–f**, Spike-generation process. **a**, Place-input-driven firing rates were determined by the distance between the animal and each place cell's field center. **b**, This distance was converted to a firing rate using a Gaussian tuning curve. **c**, Pre-synaptic input-driven firing rates were computed from the weighted sum of pre-synaptic activity. **d**, The weighted sum was passed through a sigmoid nonlinearity to obtain the final rate. **e**, CA3 cell firing rates were computed as the sum of place-input-generated and pre-synaptic-input-generated components. **f**, CA1 cell firing rates were determined solely by pre-synaptic-input-generated activity. **g**, Behavioral time-scale synaptic plasticity (BTSP). Each pre-synaptic action potential produced an eligibility trace (ET), and each post-synaptic dendritic plateau potential (PP) generated an instructive signal (IS). Synaptic weight updates depended on the product of ET and IS. **h**, Modulation of PP probability. In our model, the probability of PP occurrence was computed as the product of a neuron's recent activity, population-level normalization, and the perceived salience of the current state.

2 Pseudo code of the online learning

Algorithm 1: Training during online maze exploration

Input : $\mathcal{S}_{\text{lap}}, I_s, P_s, \text{MI}$

Output: $W_{\text{CA3}}, W_{\text{CA1}}, W_{\text{pred}}$

Initialize:

```
# Initialize weights and signals.
 $W_{\text{CA3}} \sim \mathcal{N}(10^{-2}, 10^{-8}); W_{\text{CA1}} \sim \mathcal{N}(10^{-2}, 10^{-8}); W_{\text{pred}} \leftarrow 0$ 
 $A_{\text{CA3}} \leftarrow 0; \text{ET}_{\text{CA3}} \leftarrow 0; \text{IS}_{\text{CA3}} \leftarrow 0; \text{IS}_{\text{CA1}} \leftarrow 0$ 
```

for lap = 1 **to** Totallaps **do**

for $s \in \mathcal{S}_{\text{lap}}$ **do**

for $t = 1$ **to** T_s **do**

 # Generate new spike trains and calculate the activities at every $T_{\text{SpikeUpdate}}$.

if $\text{mod}(t, T_{\text{SpikeUpdate}}) = 0$ **then**

$\text{SpikeTrain}_{\text{CA3}} \leftarrow \text{GenerateSpikeTrain}(s, A_{\text{CA3}}, W_{\text{CA3}})$
 $A_{\text{CA3}} \leftarrow \text{Activity}(\text{SpikeTrain}_{\text{CA3}})$

$\text{SpikeTrain}_{\text{CA1}} \leftarrow \text{GenerateSpikeTrain}(\cdot, A_{\text{CA3}}, W_{\text{CA1}})$
 $A_{\text{CA1}} \leftarrow \text{Activity}(\text{SpikeTrain}_{\text{CA1}})$

end

 # Compute novelty and train the feature prediction network.

$(N(t), W_{\text{pred}}) \leftarrow \text{DeltaRule}(P_s, A_{\text{CA1}}, W_{\text{pred}})$
 $\text{PS}(t) \leftarrow \text{PerceivedSaliency}(I_s, N(t), \text{MI})$

 # Calculate the plateau probability of hippocampal neurons.

$p_{\text{plateau}, \text{CA3}} \leftarrow \text{PlateauProbability}(A_{\text{CA3}}, \cdot)$
 $p_{\text{plateau}, \text{CA1}} \leftarrow \text{PlateauProbability}(A_{\text{CA1}}, \text{PS}(t))$

 # Update ETs and ISs.

$\text{ET}_{\text{CA3}} += \text{UpdateET}(\text{SpikeTrain}_{\text{CA3}})$
 $\text{IS}_{\text{CA3}} += \text{UpdateIS}(p_{\text{plateau}, \text{CA3}})$
 $\text{IS}_{\text{CA1}} += \text{UpdateIS}(p_{\text{plateau}, \text{CA1}})$

 # Train the hippocampal network.

$W_{\text{CA3}} \leftarrow \text{BTSP}(\text{ET}_{\text{CA3}}, \text{IS}_{\text{CA3}}, W_{\text{CA3}})$
 $W_{\text{CA1}} \leftarrow \text{BTSP}(\text{ET}_{\text{CA3}}, \text{IS}_{\text{CA1}}, W_{\text{CA1}})$

 # Decay eligibility traces and instructive signals.

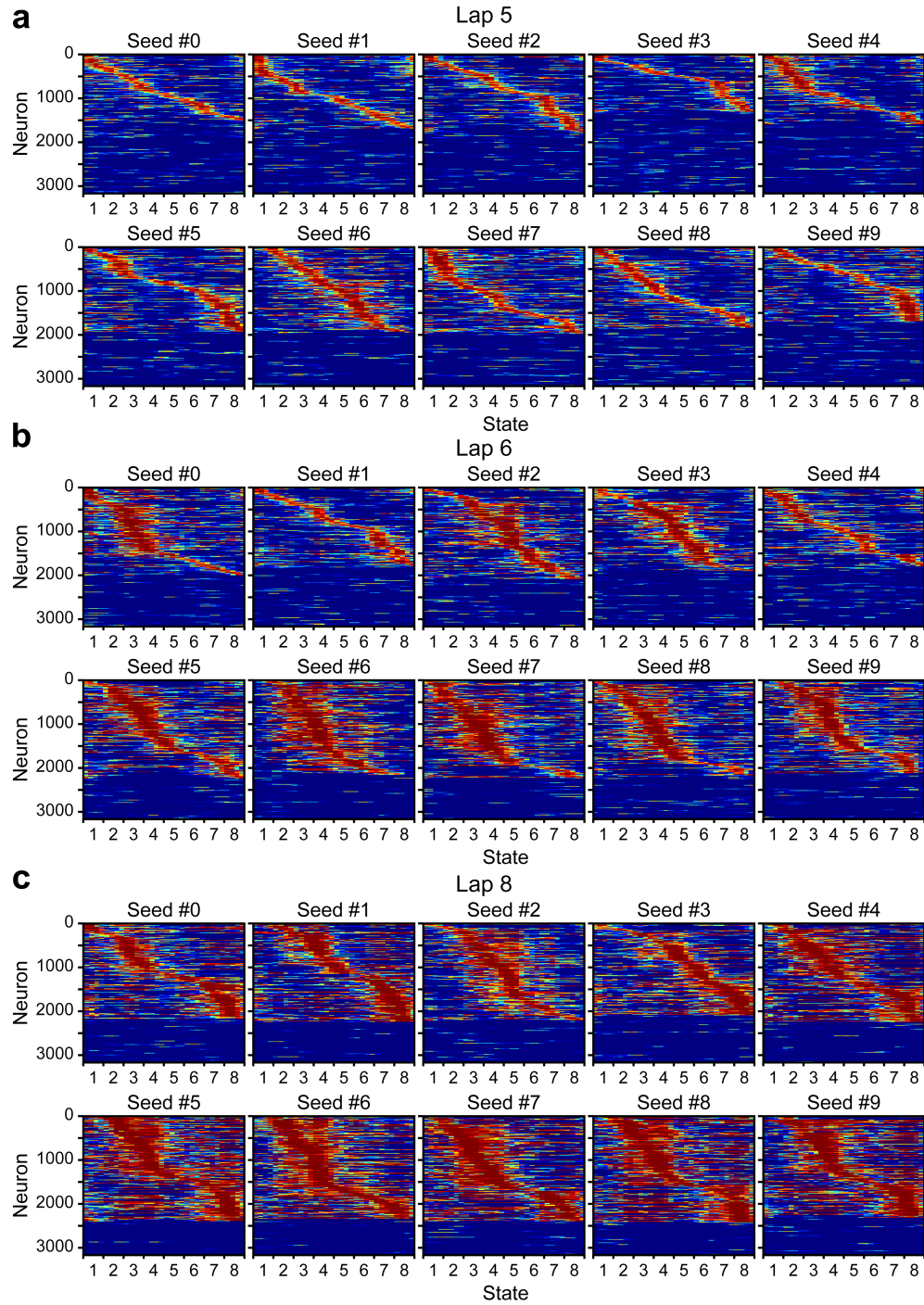
$\text{ET}_{\text{CA3}} -= \text{ET}_{\text{CA3}} / \tau_{\text{CA3}}$
 $\text{IS}_{\text{CA3}} -= \text{IS}_{\text{CA3}} / \tau_{\text{CA3}}$
 $\text{IS}_{\text{CA1}} -= \text{IS}_{\text{CA1}} / \tau_{\text{CA1}}$

end

end

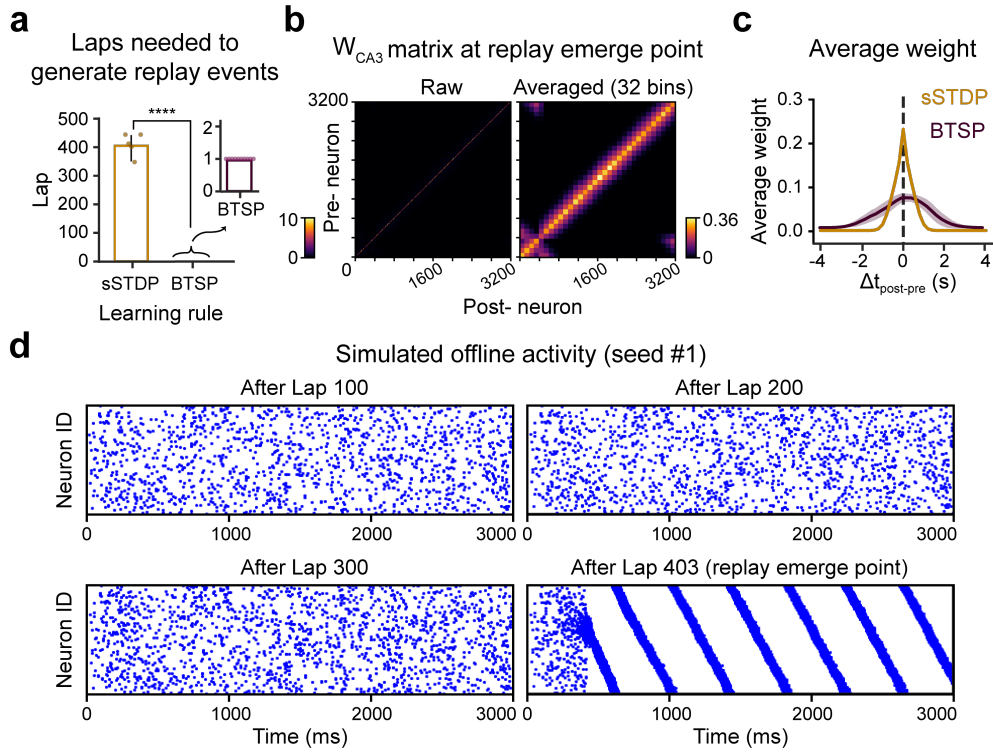
end

3 CA1 place cell activity of individual seeds



Supplementary Figure 2: CA1 cell activity across random seeds after different numbers of laps in the linear-treadmill simulations (related to Fig. ??-??). **a**, Activity after lap 5. **b**, Activity after lap 6. **c**, Activity after lap 8.

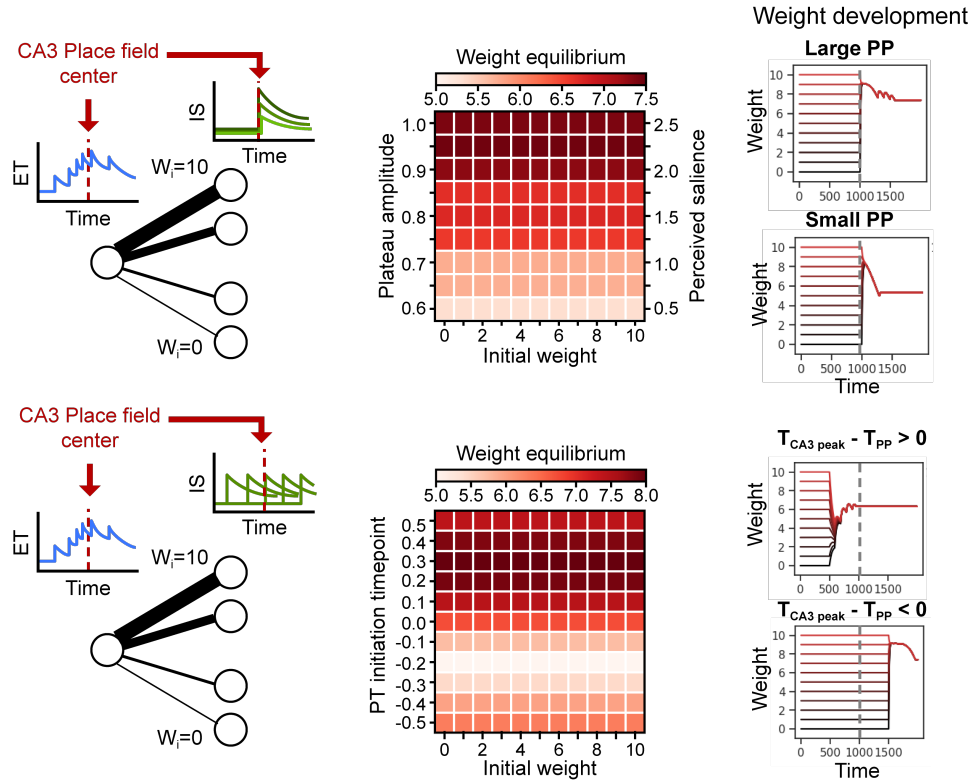
4 Online learning using symmetric STDP



Supplementary Figure 3: Laps the animal needs to travel for its CA3 network to acquire spontaneously-generated replay activity, in the linear-treadmill simulations. **a**, Replay emerge point, defined by the number of laps needed to generate spontaneous replays. The networks trained with symmetric STDP (sSTDP) demonstrated much later emergence of replay, compared to one-shot acquisition of replay within the networks trained using BTSP. **b**, W_{CA3} of a network trained using sSTDP at the replay emerge point. Left: W_{CA3} , Right: Averaged W_{CA3} computed by dividing the matrix into 32 bins and computed, plotted averaged value of each bin. **c**, Average connection weight profile of networks trained by sSTDP and BTSP. sSTDP trained weights showed stronger connection weight to other CA3 place cells with nearby place fields. BTSP trained weights are projected more broadly, projecting weights to place cells with far place fields and less strong connections to cells with nearby place fields. **d**, Simulated offline phase of sSTDP trained network. Before the replay emerge point, the activity does not show significant clustered firing or replay like activity. At lap 403 which is its replay emerge point, replay activity emerge.

5 Equilibrium properties of BTSP

Equilibrium value of a weight does not depend on the initial value of it



Supplementary Figure 4: **Laps required for CA3 networks to acquire spontaneous replay activity in linear-treadmill simulations.** **a**, The replay-emergence point, defined as the number of laps required before spontaneous replay is first observed. Networks trained with symmetric STDP (sSTDP) required substantially more experience to generate replay, whereas BTSP enabled one-shot acquisition of replay after a single lap. **b**, Example W_{CA3} from an sSTDP-trained network at its replay-emergence point. Left: full weight matrix. Right: binned and averaged weight profile (32 bins). **c**, Average CA3 connection-weight profiles for sSTDP- and BTSP-trained networks. sSTDP produced strong, narrowly tuned connections between place cells with nearby fields, whereas BTSP produced broader, more distributed connectivity projecting to distant place fields. **d**, Offline-phase activity of an sSTDP-trained network. Before the replay-emergence point, spontaneous activity showed no structured sequences. At lap 403—the replay-emergence point—robust replay events appeared.

6 Offline phase simulation details

Offline-phase simulations were performed using the synaptic weight matrices W_{CA3} and W_{CA1} obtained from the online learning phase. Starting from these learned weights, we adopted the offline simulation framework of Ecker et al.[1], using their publicly available code as the basis for our implementation.

6.1 Single cell model

We used a neuron model with the adaptive exponential integrate-and-fire (AdExIF) equations [2, 3], inspired by the previous work [1]. Here, the membrane potential $V(t)$ and adaptation current $w(t)$ interact, following the equations:

$$C_m \frac{dV(t)}{dt} = -g_L(V(t) - V_{\text{rest}}) + g_L \Delta_T \exp\left(\frac{V(t) - \vartheta}{\Delta_T}\right) - w(t) + I_{\text{syn}}(t), \quad (1)$$

$$\tau_w \frac{dw(t)}{dt} = a(V(t) - V_{\text{rest}}) - w(t). \quad (2)$$

When $V(t)$ crosses the threshold ϑ , it is reset to V_{reset} and held for a refractory period t_{ref} . The model incorporates adaptation process, which is important in replay generation [1]. The adaptation current $w(t)$ is incremented by b at each spike, where a and b control subthreshold and spike-triggered adaptation, respectively.

The actual parameter values used for the simulations are summarized in Table 1, largely adapted from the previous work [1]. Physical units are as follows: C_m (pF), g_L and a (nS), V_{rest} , Δ_T , ϑ , and V_{reset} (mV), t_{ref} and τ_w (ms), and b (pA).

Table 1: Neuron model parameters.

Parameter	Symbol	Value
Membrane capacitance	C_m	180.13
Leak conductance	g_L	5
Leak reversal (resting) potential	V_{rest}	-75.19
Slope of exponential term	Δ_T	4.23
Firing threshold	ϑ	-24.42
Adaptation time constant	τ_w	84.93
Subthreshold adaptation strength	a	-0.01
Spike-triggered adaptation increment	b	500
Reset potential	V_{reset}	-29.74
Absolute refractory period	t_{ref}	5.96

6.2 Synapse model

Synapses were modeled as conductances with biexponential kinetics:

$$g(t) = \hat{g}A \left[\exp\left(-\frac{t}{\tau_d}\right) - \exp\left(-\frac{t}{\tau_r}\right) \right], \quad (3)$$

where $\hat{g}$ is the peak conductance, and τ_r and τ_d are the rise and decay time constants, respectively. The normalization constant $A = \exp(-t_p/\tau_d) - \exp(-t_p/\tau_r)$ ensures that the conductance peaks at time $t_p = \tau_r \tau_d / (\tau_d - \tau_r) \ln(\tau_d/\tau_r)$. Synaptic currents were computed as

$$I_{\text{syn}}(t) = g_{\text{AMPA}}(t) (V(t) - E_{\text{exc}}) + g_{\text{GABA}}(t) (V(t) - E_{\text{inh}}), \quad (4)$$

where $E_{\text{exc}} = 0$ mV and $E_{\text{inh}} = -70$ mV are the reversal potentials of excitatory and inhibitory synapses, respectively.

The parameters used for the synapses are summarized in Table 2. Ext stands for excitatory cells, Inh stands for inhibitory cells, and EC stands for entorhinal cortex populations. Physical dimensions are as follows: $\hat{g}$ (nS), τ_r , τ_d , and t_d (ms), and connection probability p_{conn} is dimensionless.

Table 2: Synaptic parameters (adapted from [?, ?, ?, ?, ?]).

Connection	$\hat{g}$ (nS)	τ_r (ms)	τ_d (ms)	t_d (ms)	p_{conn}
Ext $\rightarrow$ Ext	0.1-6.3	0-15	1.3	9.5	0.1
Ext $\rightarrow$ Inh	0.85	1.0	4.1	0.9	0.1
Inh $\rightarrow$ Ext	0.65	0.3	3.3	1.1	0.25
Inh $\rightarrow$ Inh	5.0	0.25	1.2	0.6	0.25
EC $\rightarrow$ Ext	19.15	21.5	5.4	–	–

6.3 Connection weight optimization

Synaptic weights of the network were optimized using the publicly available code from the previous study [1] after being modified to incorporate W_{CA1} network. We sought to determine an optimal set of single-valued parameters determining the strength of excitatory and inhibitory connections within and between hippocampal subfields. Specifically, the parameters optimized are listed below:

- Entorhinal cortex (EC) input rate
- EC $\rightarrow$ CA3 excitatory connection weight
- CA3 excitatory $\rightarrow$ CA3 inhibitory connection weight
- CA3 inhibitory $\rightarrow$ CA3 excitatory connection weight
- CA3 inhibitory $\rightarrow$ CA3 inhibitory connection weight
- Weight scaling factor for W_{CA3}
- CA1 excitatory $\rightarrow$ CA1 inhibitory connection weight
- CA1 inhibitory $\rightarrow$ CA1 excitatory connection weight
- CA1 inhibitory $\rightarrow$ CA1 inhibitory connection weight

Optimization followed the evolutionary multi-objective framework [1], with customized upper and lower bounds for each parameter listed in Table 3. The final optimized weights used in the three task environments are summarized in Table 4.

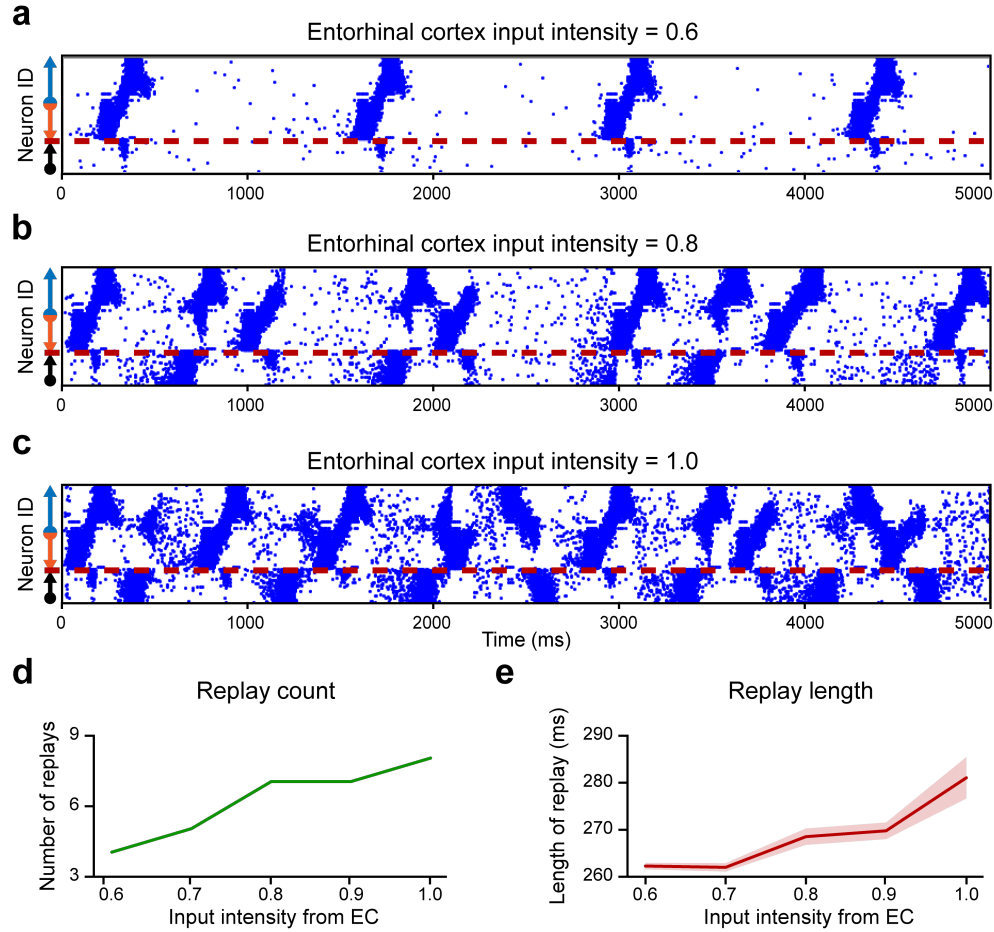
Table 3: Upper bound and lower bound values used for the optimization process.

Parameter	Lower bound	Upper bound
EC input rate	5.0	15.0
EC $\rightarrow$ CA3 Ext	0.5	40.0
CA3 Exc $\rightarrow$ Inh	0.2	5.0
CA3 Inh $\rightarrow$ Exc	0.5	20.0
CA3 Inh $\rightarrow$ Inh	1.0	15.0
CA3 weight scale (W_{CA3})	1.0	2.5
CA1 Exc $\rightarrow$ Inh	0.2	5.0
CA1 Inh $\rightarrow$ Exc	1.0	10.0
CA1 Inh $\rightarrow$ Inh	1.0	15.0

Table 4: Optimized connection weights. Each value represents the final parameter obtained after evolutionary optimization.

Parameter	Linear treadmill	T-maze	Closed linear track
EC input rate	10.084	7.322	13.038
EC $\rightarrow$ CA3 Ext	23.957	34.481	15.843
CA3 Exc $\rightarrow$ Inh	0.349	1.765	1.715
CA3 Inh $\rightarrow$ Exc	1.941	1.381	1.646
CA3 Inh $\rightarrow$ Inh	2.971	11.110	14.732
CA3 weight scale (W_{CA3})	2.210	1.886	2.306
CA1 Exc $\rightarrow$ Inh	1.000	1.765	0.337
CA1 Inh $\rightarrow$ Exc	1.198	1.877	1.052
CA1 Inh $\rightarrow$ Inh	14.725	13.819	12.671

7 Replay frequency modulation based on perceived salience



Supplementary Figure 5: Replay frequency and event length are modulated by the strength of entorhinal cortex (EC) input. **a–c**, Simulated offline activity in networks that learned the T-maze task under different EC input strengths (related to Fig. ??–??). **a**, EC input at 60% of the standard strength. **b**, EC input at 80% of the standard strength. **c**, EC input at 100% (full) strength. **d**, Number of replay events as a function of EC input strength. **e**, Replay event length as a function of EC input strength.

References

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